

Examples based on List and dictionary comprehension:

1. Write a list comprehension to generate a list of squares of numbers from 1 to 10.
2. Use a list comprehension to extract all even numbers from the list .
3. Create a list of squares of only even numbers from 1 to 10 using a single list comprehension.
4. Given a list of strings:
`words = ["apple", "banana", "cherry", "date"]`
Write a list comprehension to create a list of their lengths.
5. Write a list comprehension to generate a list of all odd numbers between 1 and 20.
6. Generate a list of the first 10 multiples of 7 using list comprehension.
7. Given a list of strings:
`names = ["ALICE", "BOB", "CHARLIE"]`
Use a list comprehension to convert all strings to lowercase.
8. From the list `words = ["python", "java", "c++", "html", "css"]`, use a list comprehension to create a list of words that have more than 4 characters.
9. Given two lists:
`list1 = [1, 2, 3, 4]` and `list2 = [3, 4, 5, 6]`
Use a list comprehension to find the common elements.
10. Given a list of words:
`words = ["hello", "world", "python"]`
Create a list comprehension to reverse each string.
11. Write a list comprehension to compute the square root of numbers in the list:
`nums = [4, 16, 25, 36]`.
12. Create a dictionary where the keys are numbers from 1 to 5, and the values are their cubes.
13. Given a dictionary:
`nums = {'a': 1, 'b': 2, 'c': 3, 'd': 4}`
Use a dictionary comprehension to create a new dictionary with only the even values squared.
14. Given a list:
`fruits = ["apple", "banana", "pear"]`
Create a dictionary mapping each fruit to its length.
15. Write a dictionary comprehension to count the frequency of each character in the string:
`text = "hello"`
16.

From the dictionary:
`numbers = {'one': 1, 'two': 2, 'three': 3, 'four': 4}`
Use a dictionary comprehension to create a new dictionary with keys that map to "odd" or "even" based on their values.
17. Create a dictionary comprehension to map each character in the string "abcde" to its ASCII value.
18. Given a dictionary:
`values = {'a': -1, 'b': 2, 'c': -3, 'd': 4}`
Use a dictionary comprehension to filter out keys with negative values.
19. Create a dictionary where the keys are numbers from 1 to 5, and the values are their cubes.

20. Given a dictionary:

```
nums = {'a': 1, 'b': 2, 'c': 3, 'd': 4}
```

Use a dictionary comprehension to create a new dictionary with only the even values squared.

21. Given a list:

```
fruits = ["apple", "banana", "pear"]
```

Create a dictionary mapping each fruit to its length.